

ASHUTOSH ROY

AI Engineer | AI Agents & LLM Specialist

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SUMMARY

AI Engineer specializing in autonomous AI agents, multi-agent orchestration, and large language model (LLM) fine-tuning. Expertise in LangChain, LangGraph, and integrating Model Context Protocol (MCP) servers for robust, tool-augmented agentic workflows. Published researcher with hands-on experience in building hybrid retrieval pipelines, advanced RAG architectures, and deploying production-ready generative AI systems via FastAPI.

EXPERIENCE

Research Intern

Jan 2025 – Jun 2025

Indian Institute of Technology (IIT) Delhi

New Delhi, India

- Designed hybrid retrieval pipelines (FAISS/Pinecone + Apache Solr) optimizing tool usage for document-querying AI agents over complex archaeological datasets.
- Deployed scalable **FastAPI** services for real-time LLM inference and function calling; reducing retrieval-augmented generation (RAG) latency by 35%.
- Co-authored **SARCH: Multimodal Search for Archaeological Archives** with IIT Delhi collaborators — submitted for peer review. arxiv.org/abs/2511.05667

AI/ML Engineer Intern

Oct 2024 – Nov 2024

MitoVoid AI

Remote

- Fine-tuned transformer-based models (Mistral/LLaMA) via QLoRA for healthcare domain, enabling a conversational AI agent to handle medical record summarization and symptom analysis.
- Integrated autonomous tool-calling capabilities to patient inquiry bots, reducing query resolution time by ~40% through optimized intent routing.
- Built end-to-end ML pipelines bridging data preprocessing, model inference, and real-time interactive deployment.

PROJECTS

Shopping Assistant Agent (LangGraph & MCP) | Python, LangGraph, MCP, Claude 3.5, FastAPI

Feb 2026 - May 2026

- Engineered a multi-agent shopping assistant using **LangGraph** to coordinate specialized agents for product search, comparison, recommendation, and review summarization with self-reflection and retry loops.
- Implemented custom **Model Context Protocol (MCP)** servers enabling secure real-time access to product APIs, shopping databases, pricing tools, and external e-commerce sources for contextual recommendations.

Skills for Agents Framework | Open-source, Python, AI-Agents

April 2026 - May 2026

- Developed a production-ready ecosystem of composable instruction sets, allowing AI agents to handle multiple tasks concurrently without prompt collision.
- Architected the **Skills Interoperability Protocol (SIP)**, dividing capabilities into domains (voice, density, craft) enabling complex layered and sequential pipeline agent behaviors.

Exam-Helper Agentic RAG | Python, LLaMA 3, LangGraph, FAISS, Pinecone

Dec 2024 - Apr 2025

- Built an LLM-powered autonomous contextual QA agent on **LLaMA 3** using LangGraph for multi-step reasoning across PDFs, images, and YouTube transcripts.
- Implemented dense retrieval (FAISS) with semantic chunking and dynamic context augmentation to drastically reduce hallucination in domain-specific academic queries.

Speech Emotion & Stress Detection | PyTorch, Wav2Vec2, BiLSTM, Hugging Face | RECCAP 2026

Aug 2024 - Dec 2024

- Proposed a self-supervised multitask architecture for simultaneous emotion classification and stress regression; **accepted at RECCAP 2026, IIT Palakkad.**

Medical Chatbot (LLM Fine-Tuning) | Python, Mistral 7B, QLoRA, Tesseract OCR

Mar 2024 - Apr 2024

- Fine-tuned **Mistral 7B** using **QLoRA** for domain-specific medical NLU, optimizing inference speed while preserving high intent-recognition accuracy.

TECHNICAL SKILLS

AI/ML & Research: LLMs, GenAI, RAG, NLP, LangChain, Transformer Fine-Tuning (QLoRA/LoRA), Self-Supervised Learning, Multimodal AI, Speech Processing, Dense Retrieval, Foundation Models

Libraries & Frameworks: PyTorch, Hugging Face, TensorFlow, Scikit-learn, FastAPI, Streamlit, LangGraph, MLOps

Databases & Search: FAISS, Pinecone, Apache Solr, MySQL

Languages & Tools: Python, C++, SQL, Git, Docker, Linux, VS Code, Google Colab

EDUCATION

Chhattisgarh Swami Vivekananda Technical University (CSVTU)

2022 – 2026

B.Tech (Hons) in Computer Science and Engineering (Artificial Intelligence)

Bhilai, Chhattisgarh

St. Columbus School

2020 – 2021

Higher Secondary (Class XII - Science PCMB) - 87%

Faridabad, Haryana